PROJECT CHARTER

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

University Student-Support Case Agent

BY GROUP M EVENING

Name	STUDENT No	REG No
Namata Sumayya	2300722615	23/U/22615
Aryampa Joshua	2300706517	23/U/06517/PS
Nakaye Hildah	2300713598	23/U/13598/PS
Mushabe Moses	2300712131	23/U/12131/EVE

Problem

Freshmen (first-year students) at Makerere frequently need answers to routine onboarding questions, that is configuring their university webmail, MUELE, connecting to campus Wi-Fi, registration requirements, fee-related processes, password resets, and other similar first-year onboarding needs, among others. Today, this information sits across different university documents, university websites, and separate departmental offices. Getting answers typically means searching several documents or physically visiting an office/queue. This creates unnecessary delay for students and repetitive, low-value workload for support staff who re-answer the same categories of questions again and again.

User

Primary user: Makerere University freshmen students (first-year students), who need quick, accurate information about onboarding tasks and early academic processes, such as configuring their webmail, connecting to Wi-Fi, registration requirements, and similar FAQs and procedures relevant to onboarding.

Secondary user (human-in-the-loop): A support-desk staff member who reviews and acts on any case the agent escalates.

Current Pain Point

Currently, when the first-year students have questions, they have to visit the related office to seek answers, and in those offices, they may have to first enter a queue and then await their turn. The support staff spends most of their time answering similar repetitive questions rather than handling genuinely complex cases that need human judgement.

AI Value

This is a strong fit for AI because the problem has two distinct parts an LLM can help with:

- Retrieval-grounded Q&A: most student questions, especially the onboarding questions from freshmen, are answerable from existing documents and the different university websites, if the right passage can be found and summarized accurately. An AI system with RAG can do this instantly, citing the source, rather than a student manually searching PDFs.
- Bounded multi-step handling: Some requests aren't just "look it up"; they need a new case to be raised rather than just answered. An agent that can retrieve relevant information, decide when a question needs escalation, and create a ticket in one flow removes friction a static FAQ or chatbot can't, while keeping every higher-impact action (approving, resolving, deciding a case) with a human.

AI is not being used here to make decisions on a freshman's behalf it's used to remove the search and routing burden, which is exactly the kind of repetitive, well-bounded task that benefits from automation while leaving judgment calls to staff.

Scope

In scope for this 8-week build

- Answering freshman onboarding questions (e.g., configuring university webmail, connecting to campus Wi-Fi, registration requirements, and similar first-year FAQs) grounded in the information that we provide for the AI, with source citation and an honest fallback message for unsupported questions.
- Creating/routing a new support case to the appropriate category.
- Remembering the active case within a session so freshmen can ask follow-up questions without repeating the case ID.
- Basic evaluation of the above against normal, edge, and adversarial test questions.
- Recognizing requests outside the agent's authority and escalating them to support-desk staff rather than attempting to resolve them.

Explicitly out of scope

- Any admissions, grading, disciplinary, or fee/financial decisions.
- Automatically resolving or closing a case; resolution stays with a human.
- Real student data or integration with actual university systems.
- General-purpose chat unrelated to student support.
- Questions from non-freshman students (continuing, returning, or postgraduate students) or any need falling outside the freshman onboarding FAQ scope described above.

Assumptions and Constraints

Data

We will use publicly available Makerere documents relevant to freshman onboarding (IT/webmail and Wi-Fi setup guides, registration guidelines, academic regulations, academic calendar, timetables available online, and any other online source material that we may find relevant) for the knowledge base, and team-created synthetic data (fake student records, fake case/ticket records) to simulate the case-management system. No real student data will be used.

Timeline

The system activities that are building, testing must be done in a period of 8 weeks by ensuring each week deliverable is done within the timeline.

Skills/tooling

The team's current strengths are in frontend/UI-UX and early AI and ML exposure, and the weekly knowledge that will be obtained in the lectures on emerging trends in software.

Access: no confidential institutional data or system access has been requested or is required because we shall use what is available to the public.

Minimum Proposal Statement

Our system helps Makerere freshmen (first-year students) find accurate answers to onboarding and support/policy questions, such as configuring university webmail, connecting to campus Wi-Fi, registration requirements, and similar first-year FAQs, and raising a support case when needed. AI is used for retrieval-grounded question answering and reasoning about when a question requires escalation versus a direct answer. Deterministic software remains responsible for writing case records, applying category rules, and validating inputs. The agent may use a document-retrieval (RAG) tool and a case-creation tool, but may not make admissions, grading, disciplinary, or fee decisions, or resolve/close a case without human approval. We will build and evaluate using a ~10–50-document corpus of public Makerere onboarding materials plus a team-created synthetic case dataset.